

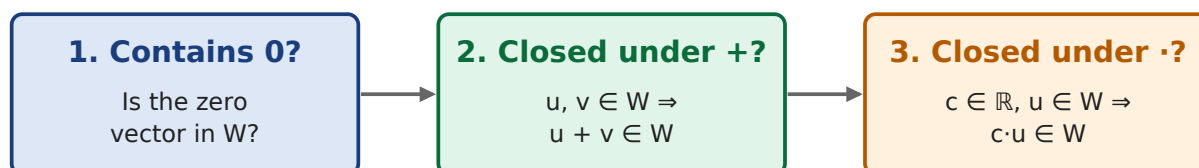
Linear Algebra — Practice Set

Vector Spaces & Subspaces • Linear Combination, Dependence & Independence • Basis & Dimension

THE BIG IDEA

Subspaces are vector spaces. Every subspace W of a vector space V is itself a vector space (under the same addition and scalar multiplication as V). Once you confirm closure and the zero vector, all eight vector-space axioms are inherited automatically — that's why the **3-step Subspace Test** is enough.

The Subspace Test



If all three pass $\rightarrow W$ is a subspace (and therefore a vector space in its own right).

DEFINITION — Vector Space

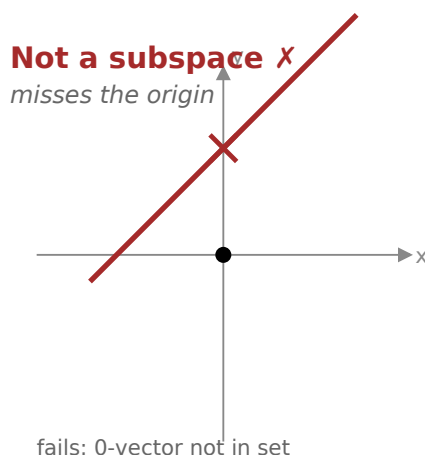
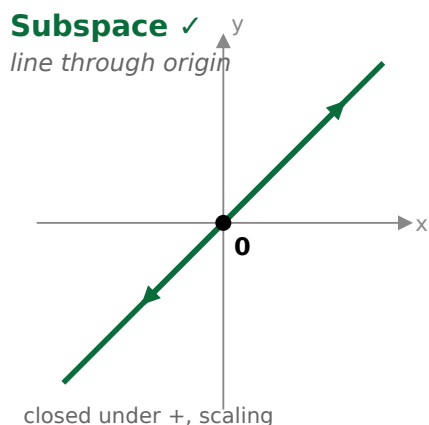
A set V together with two operations — vector addition $(+)$ and scalar multiplication $(\cdot)$ — that satisfies 8 axioms: closure under $+$ and $\cdot$, commutativity and associativity of $+$, existence of a zero vector and additive inverses, and four distributive/identity laws ($c(u+v) = cu + cv$, $(c+d)u = cu + du$, $(cd)u = c(du)$, $1 \cdot u = u$). Examples: $\mathbb{R}^n$, polynomials, $m \times n$ matrices, continuous functions on $[a, b]$.

DEFINITION — Subspace

A subset $W \subseteq V$ is a **subspace** if it is itself a vector space using the same operations as V . By the **Subspace Test**: W is a subspace $\Leftrightarrow$ (1) $0 \in W$, (2) $u, v \in W \Rightarrow u+v \in W$, (3) $c \in \mathbb{R}, u \in W \Rightarrow c \cdot u \in W$.

INTUITION

Picture a subspace as a flat sheet (a line, a plane, or higher-dimensional analogue) **passing through the origin**. You can never "escape" it by adding two of its arrows or by stretching one — it is a closed world. If a candidate set misses the origin, or you can hop out by adding/scaling, it is *not* a subspace.



How to use this worksheet: Each section opens with a short concept primer (definitions, intuition, geometric picture), followed by 6 problems graded as Easy / Medium / Difficult. Solutions are written step-by-step in plain language so students can self-check.

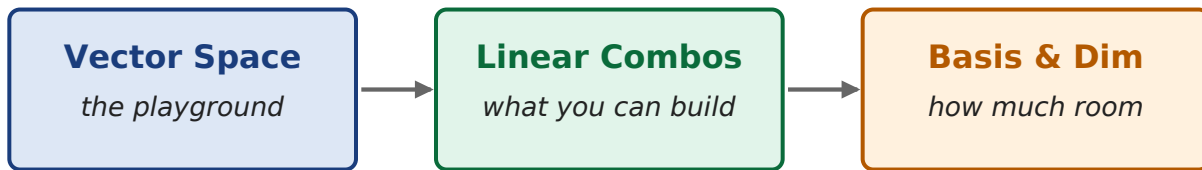
What's Inside

	Topic	Concepts you'll master	Page
§1	Vector Spaces & Subspaces	the 8 axioms, the 3-step subspace test, lines & planes through origin	4
§2	Linear Combinations, Dependence & Independence	vectors as recipes, redundancy, REF & determinant methods	7
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How the Three Topics Fit Together

These three topics are the spine of linear algebra and they build on each other. The picture below shows the dependency: **vector spaces** are the playground; **linear combinations** describe what you can build inside; **basis & dimension** measure how much room there is.

Concept flow: structure → operations → measurement



ANALOGY

A useful analogy: a vector space is like a kitchen. Linear combinations are the recipes you can cook. The basis is your minimum-shopping-list of ingredients, and the dimension is how many ingredients are on that list. Different shopping lists (bases) can cook the same menu, but they all have the same length (dimension).

Section 1 — Vector Spaces and Subspaces

INTUITION

Vectors aren't just arrows. They are anything you can add together and stretch — tuples in $\mathbb{R}^n$, polynomials, matrices, even functions. A **vector space** is the playground for these objects, and a **subspace** is a smaller playground inside that still works on its own. The magic: passing the 3-step Subspace Test gets you all 8 vector space axioms for free, because they were already true in the parent.

STRATEGY

How to attack a subspace problem:

1. **Check 0:** Plug in the zero vector — does it satisfy the rule defining W ? (If not, stop — W is not a subspace.)
 2. **Check +:** Pick generic $u, v \in W$ and verify that $u + v$ still satisfies the rule.
 3. **Check $\cdot$:** Pick generic $c \in \mathbb{R}$ and $u \in W$; verify $c \cdot u$ still satisfies the rule.
- If any one fails, give a concrete **counter-example** — that is enough to disprove.

WATCH OUT

Sets defined with **strict inequalities** (like $x > 0$), **nonzero constants** (like $x + y = 5$), or **products** of variables (like $xy = 0$) typically fail the test. Subspaces correspond to **linear, homogeneous** conditions like $x + 2y - 3z = 0$.

★ Easy

Q1 (Easy): Show that $W = \{(x, 0) : x \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.

Solution:

Strategy: apply the three subspace tests.

(1) Zero vector: Take $x = 0$. Then $(0, 0)$ is in W . ✓

(2) Closed under addition: Take $u = (a, 0)$ and $v = (b, 0)$ in W . Then $u + v = (a + b, 0)$. The second coordinate is still 0, so $u + v \in W$. ✓

(3) Closed under scalar multiplication: For $c \in \mathbb{R}$ and $u = (a, 0) \in W$, $c \cdot u = (ca, 0) \in W$. ✓

All three pass, so **W is a subspace of $\mathbb{R}^2$.**

In simple words: W is just the x -axis. A line through the origin in $\mathbb{R}^2$ is always a subspace.

Q2 (Easy): Determine whether $W = \{(x, y) \in \mathbb{R}^2 : x + y = 0\}$ is a subspace of $\mathbb{R}^2$.

Solution:

Strategy: the rule $x + y = 0$ is linear and homogeneous — promising.

(1) Zero vector: $0 + 0 = 0$, so $(0, 0) \in W$. ✓

(2) Addition: If $a + b = 0$ and $c + d = 0$, then $(a + c) + (b + d) = (a + b) + (c + d) = 0$. ✓

(3) Scalar multiplication: $ka + kb = k(a + b) = k \cdot 0 = 0$. ✓

Hence **W is a subspace of $\mathbb{R}^2$** .

Geometric picture: W is the line $y = -x$ through the origin.

★★ Medium

Q3 (Medium): Show that $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ is a subspace of $\mathbb{R}^3$.

Solution:

Strategy: linear homogeneous equation in 3 variables — likely a plane through origin.

(1) Zero: $0 + 0 + 0 = 0$. $(0,0,0) \in W$. ✓

(2) Addition: If $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$, then $(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = 0 + 0 = 0$. ✓

(3) Scaling: $c \cdot (x + y + z) = c \cdot 0 = 0$. ✓

Therefore **W is a subspace of $\mathbb{R}^3$** . Geometrically, it's a plane through the origin.

Q4 (Medium): Show that $W = \{(x, y) \in \mathbb{R}^2 : x \geq 0\}$ is NOT a subspace of $\mathbb{R}^2$.

Solution:

Strategy: only one failure is needed for disproof. Try a negative scalar.

Take $u = (2, 3) \in W$ (since $2 \geq 0$) and $c = -1$. Then $c \cdot u = (-2, -3)$, and -2 is not ≥ 0 .

So $c \cdot u \notin W$. W fails closure under scalar multiplication, hence **W is not a subspace**.

In simple words: W is the right half-plane. Negating a vector flips it to the other side, violating closure.

★★★ Difficult

Q5 (Difficult): Let V be the set of all 2×2 real matrices. Show that the set W of all 2×2 symmetric matrices (matrices A with $A^T = A$) is a subspace of V.

Solution:

Strategy: use transpose properties $(A+B)^T = A^T + B^T$ and $(cA)^T = c \cdot A^T$.

(1) Zero matrix: The 2×2 zero matrix equals its own transpose. So $0 \in W$. ✓

(2) Addition: If $A^T = A$ and $B^T = B$, then $(A+B)^T = A^T + B^T = A + B$. So $A+B \in W$. ✓

(3) Scaling: $(cA)^T = c \cdot A^T = c \cdot A$. So $cA \in W$. ✓

All three hold, so **W is a subspace of V**.

In simple words: symmetric matrices mirror across the diagonal. Adding or scaling preserves that mirror symmetry.

Q6 (Difficult): Show that the set P_n of all polynomials of degree $\leq n$ (with the zero polynomial included) is a vector space under usual polynomial addition and scalar multiplication.

Solution:

Strategy: P_n sits inside the vector space of all polynomials, so the 8 axioms (commutativity, associativity, distributivity, identity, inverses) are inherited from real addition. We just verify closure and the special elements.

Closure under addition: If $\deg(p) \leq n$ and $\deg(q) \leq n$, then $\deg(p+q) \leq n$. ✓

Closure under scaling: If $\deg(p) \leq n$ and $c \in \mathbb{R}$, then $\deg(c \cdot p) \leq n$. ✓

Zero element: the zero polynomial $0(x)$ is in P_n and $p + 0 = p$. ✓

Additive inverse: for $p(x) = a_0 + a_1x + \dots + a_nx^n$, $-p(x) = -a_0 - a_1x - \dots - a_nx^n \in P_n$ and $p + (-p) = 0$. ✓

Remaining axioms: commutativity, associativity, distributivity ($(c+d)p = cp + dp$, $c(p+q) = cp + cq$), associativity of scalar product $(cd)p = c(dp)$, and identity $1 \cdot p = p$ — all follow coefficient-wise from properties of real numbers. ✓

All 8 axioms hold, so **P_n is a vector space.**

In simple words: polynomials behave like vectors — coefficients act like coordinates.

Section 2 — Linear Combination, Dependence and Independence

DEFINITION — Linear Combination

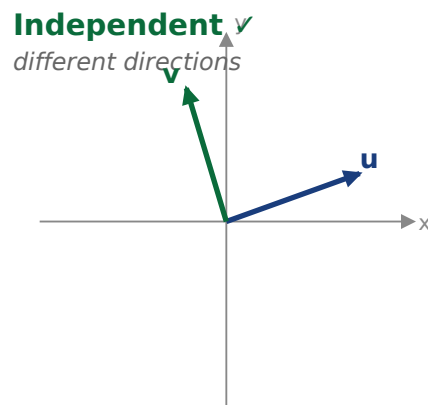
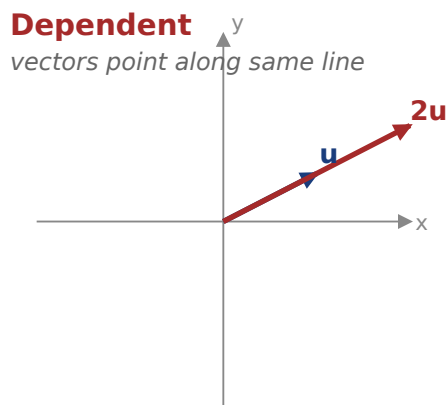
A vector v is a **linear combination** of $v_1, v_2, \dots, v_k$ if there exist scalars $c_1, \dots, c_k \in \mathbb{R}$ with $v = c_1v_1 + c_2v_2 + \dots + c_kv_k$.

DEFINITION — Linear Independence

Vectors $v_1, \dots, v_k$ are **linearly independent** if the only way to make $c_1v_1 + \dots + c_kv_k = 0$ is $c_1 = c_2 = \dots = c_k = 0$ (the trivial solution). Otherwise they are **linearly dependent** — at least one vector is redundant (it can be written using the others).

INTUITION

Linear combination = a recipe. Vectors are the ingredients; scalars are how much of each. **Independence** = no ingredient is wasted: you can't make any of them from the others. **Dependence** = duplicate work — at least one vector lies in the span of the others, so removing it doesn't shrink what you can build.



STRATEGY

Independence test (the standard recipe):

1. Set $c_1v_1 + c_2v_2 + \dots + c_kv_k = 0$.
2. Translate into a homogeneous linear system (one equation per coordinate).
3. Solve. If only $c_1 = \dots = c_k = 0$ works $\rightarrow$ independent. If a non-trivial solution exists $\rightarrow$ dependent.

Shortcut for n vectors in $\mathbb{R}^n$: form the $n \times n$ matrix and check $\det \neq 0 \Leftrightarrow$ independent.

WATCH OUT

Two vectors are dependent $\Leftrightarrow$ one is a scalar multiple of the other. But for three or more vectors, dependence does NOT mean two of them are parallel — it just means one is in the span of the others. Always set up the equation and solve.

THE REF METHOD

Row Echelon Form (REF) is the unified workhorse for these problems. Form a matrix from the vectors (as rows or columns), reduce to REF, and read off the answer:

- **Linear combination** — augment with the target vector, reduce, back-substitute.
- **Independence** — count the non-zero rows; if it equals the number of vectors, they are independent (rank = number of vectors).
- **Span / basis** — the non-zero rows of REF give a basis for the row space.

★ **Easy**

Q1 (Easy): Express $v = (3, 7)$ as a linear combination of $u_1 = (1, 1)$ and $u_2 = (1, 2)$.

Solution:

Strategy: set $v = a \cdot u_1 + b \cdot u_2$ and solve for a, b coordinate-wise.

System: $a + b = 3, \quad a + 2b = 7.$

Step 1: Subtract row 1 from row 2: $b = 4.$

Step 2: Plug back: $a + 4 = 3 \Rightarrow a = -1.$

Answer: $v = -1 \cdot u_1 + 4 \cdot u_2.$ **Check:** $-1(1,1) + 4(1,2) = (3, 7).$ ✓

Q2 (Easy): Are the vectors $(1, 2)$ and $(2, 4)$ linearly independent in $\mathbb{R}^2$?

Solution:

Strategy: two vectors — check if one is a scalar multiple of the other.

Notice $(2, 4) = 2 \cdot (1, 2)$. So $2 \cdot (1,2) + (-1) \cdot (2,4) = (0,0)$ with non-zero scalars.

A non-trivial combination yields zero, so the vectors are **linearly dependent**.

Picture: both arrows point along the same line, so they encode the same direction.

★★ **Medium**

Q3 (Medium): Express $v = (1, -2, 5)$ as a linear combination of $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 3)$, $u_3 = (2, -1, 1)$.

Solution:

Strategy: set up a 3×3 system from $a \cdot u_1 + b \cdot u_2 + c \cdot u_3 = v$ and solve.

(i) $a + b + 2c = 1$

(ii) $a + 2b - c = -2$

(iii) $a + 3b + c = 5$

Step 1: (ii) – (i): $b - 3c = -3$... (iv)

Step 2: (iii) – (i): $2b - c = 4$... (v)

Step 3: From (iv): $b = 3c - 3$. Sub into (v): $2(3c - 3) - c = 4 \Rightarrow 5c = 10 \Rightarrow c = 2$.

Step 4: $b = 3(2) - 3 = 3$. From (i): $a = 1 - 3 - 4 = -6$.

Answer: $v = -6 \cdot u_1 + 3 \cdot u_2 + 2 \cdot u_3$.

Check: $-6(1,1,1) + 3(1,2,3) + 2(2,-1,1) = (-6,-6,-6) + (3,6,9) + (4,-2,2) = (1,-2,5)$. ✓

Alternate Solution — Augmented Matrix → REF:

Place u_1, u_2, u_3 as *columns* and v on the right to form the augmented matrix $[u_1 \mid u_2 \mid u_3 \mid v]$:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 1 & 2 & -1 & | & -2 \\ 1 & 3 & 1 & | & 5 \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & -3 & | & -3 \\ 0 & 2 & -1 & | & 4 \end{bmatrix}$$

$R_3 \rightarrow R_3 - 2R_2$:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & -3 & | & -3 \\ 0 & 0 & 5 & | & 10 \end{bmatrix}$$

Back-substitute: $5c = 10 \Rightarrow c = 2$; $b - 3c = -3 \Rightarrow b = 3$; $a + b + 2c = 1 \Rightarrow a = -6$. Same answer. ✓

Q4 (Medium): Are the vectors $(1, 0, 1)$, $(0, 1, 1)$, $(1, 1, 0)$ linearly independent in $\mathbb{R}^3$?

Solution:

Strategy: set $a \cdot v_1 + b \cdot v_2 + c \cdot v_3 = 0$ and solve the homogeneous system.

$$a + c = 0$$

$$b + c = 0$$

$$a + b = 0$$

Step 1: From row 1, $a = -c$. From row 2, $b = -c$.

Step 2: Plug into row 3: $-c + (-c) = 0 \Rightarrow -2c = 0 \Rightarrow c = 0$. Then $a = b = 0$.

Only the trivial solution works, so the vectors are **linearly independent**. ✓

Alternate Solution — Row Reduce to REF:

Form the matrix with the vectors as rows and reduce:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$R_3 \rightarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$R_3 \rightarrow R_3 - R_2$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

Three non-zero rows $\Rightarrow$ **rank = 3**. Since rank equals the number of vectors, they are **linearly independent**. ✓

Determinant shortcut: det of the matrix with these rows is $-2 \neq 0$, confirming independence.

★★★ Difficult

Q5 (Difficult): Are the polynomials $p_1(x) = 1 + x$, $p_2(x) = x + x^2$, $p_3(x) = 1 + x^2$ linearly independent in P_2 ?

Solution:

Strategy: equate a linear combination to the zero polynomial; coefficients of every power of x must vanish.

$$a(1+x) + b(x+x^2) + c(1+x^2) = (a+c) + (a+b)x + (b+c)x^2 = 0.$$

Coefficient equations:

$$(i) a + c = 0 \quad (ii) a + b = 0 \quad (iii) b + c = 0$$

Step 1: From (i), $c = -a$. From (ii), $b = -a$.

Step 2: Sub into (iii): $-a + (-a) = 0 \Rightarrow -2a = 0 \Rightarrow a = 0$. Then $b = c = 0$.

Only trivial solution $\rightarrow$ **p_1, p_2, p_3 are linearly independent.**

In simple words: none of these polynomials can be built from the other two by adding or scaling alone.

Q6 (Difficult): Determine whether $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ is linearly independent in $\mathbb{R}^3$.

Solution:

Strategy: reduce the augmented matrix; if a row vanishes, you'll get a free parameter and dependence.

$$\text{System: } a + 4b + 7c = 0, \quad 2a + 5b + 8c = 0, \quad 3a + 6b + 9c = 0.$$

Step 1: $R_2 \rightarrow R_2 - 2R_1$: $-3b - 6c = 0 \Rightarrow b + 2c = 0$.

Step 2: $R_3 \rightarrow R_3 - 3R_1$: $-6b - 12c = 0 \Rightarrow b + 2c = 0$ (same equation).

Step 3: Only two independent equations remain. Let $c = t$ (free); then $b = -2t$ and $a = -4(-2t) - 7t = t$. Pick $t = 1$: $(a, b, c) = (1, -2, 1)$.

Verify: $1 \cdot (1, 2, 3) - 2 \cdot (4, 5, 6) + 1 \cdot (7, 8, 9) = (1, 2, 3) - (8, 10, 12) + (7, 8, 9) = (0, 0, 0)$. ✓

A non-trivial combination gives zero, so the vectors are **linearly dependent**.

Alternate Solution — Row Reduce to REF:

Form the matrix with the vectors as rows and reduce to row echelon form:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 4R_1, \quad R_3 \rightarrow R_3 - 7R_1:$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2:$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

This is REF. There are only **2 non-zero rows**, so **rank = 2 < 3** (the number of vectors). Hence the vectors are **linearly dependent**.

Bonus — extracting the relation from REF: track the operations on R_3 . We did $R_3 \rightarrow R_3 - 7R_1$ then $R_3 \rightarrow R_3 - 2R_2'$ (where $R_2' = R_2 - 4R_1$). Combining: final $R_3 = (v_3 - 7v_1) - 2(v_2 - 4v_1) = v_3 + v_1 - 2v_2$. Setting this = 0 gives **$v_1 - 2v_2 + v_3 = 0$** , the same dependency relation we found by direct solving.

Quick determinant check: $\det = 0$, instantly confirming dependence.

Section 3 — Basis and Dimension

DEFINITION — Span

The **span** of vectors $v_1, \dots, v_k$ is the set of *all* their linear combinations: $\text{Span}(v_1, \dots, v_k) = \{c_1 v_1 + \dots + c_k v_k : c_i \in \mathbb{R}\}$. It is automatically a subspace of V .

DEFINITION — Basis

A **basis** of a vector space V is an ordered set $B = \{v_1, \dots, v_n\}$ that is (1) linearly independent and (2) spans V . Every vector in V then has a *unique* expression as $c_1 v_1 + \dots + c_n v_n$.

DEFINITION — Dimension

The **dimension** of V is the number of vectors in any basis (this number is the same for every basis). $\dim(\mathbb{R}^n) = n$, $\dim(P_n) = n+1$, $\dim(M_{m \times n}(\mathbb{R})) = m \cdot n$.

INTUITION

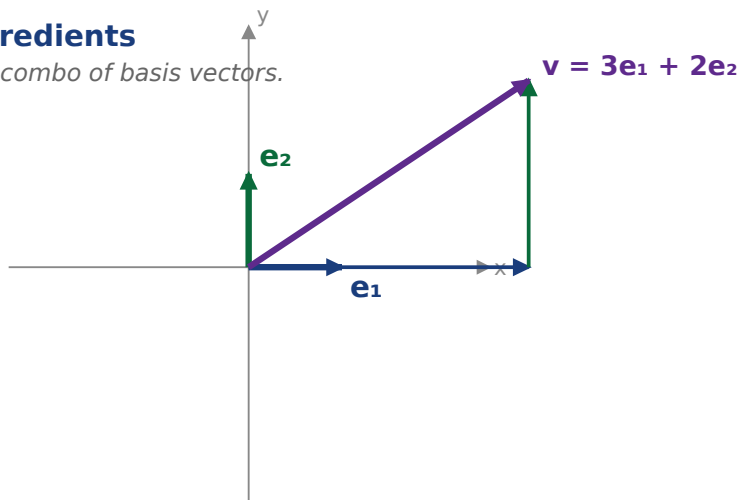
Basis = minimum complete recipe. Just enough ingredients to make every vector — no extras (independence) and nothing missing (spanning). **Dimension = degrees of freedom** — how many independent knobs you can tune. $\mathbb{R}^2$ has 2 knobs (x and y); the plane $x + y + z = 0$ in $\mathbb{R}^3$ lets you choose y and z freely, then x is forced — that's 2 degrees of freedom, $\dim = 2$.

Dimension ladder — how much room is in there?



Basis = recipe ingredients

Every vector is a unique combo of basis vectors.

**STRATEGY****Finding a basis of a subspace defined by equations:**

1. Solve the constraints — express the dependent variables in terms of the free ones.
2. Write a typical element as a sum: $(\text{free}_1) \cdot v_1 + (\text{free}_2) \cdot v_2 + \dots$
3. The vectors $v_1, v_2, \dots$ so produced span the subspace and are automatically independent.
4. The number of free variables = dimension.

HANDY SHORTCUT

Counting trick: in an n -dimensional space, *any* n linearly independent vectors form a basis automatically — you don't have to re-prove spanning. Likewise, any spanning set of size n is automatically independent. This saves a lot of work.

★ **Easy**

Q1 (Easy): Find the dimension of $\mathbb{R}^3$ and give a basis.

Solution:

Strategy: use the standard basis e_1, e_2, e_3 .

Let $e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$.

Spanning: any $(x,y,z) = x \cdot e_1 + y \cdot e_2 + z \cdot e_3$.

Independence: $a \cdot e_1 + b \cdot e_2 + c \cdot e_3 = (a,b,c) = 0$ forces $a = b = c = 0$.

So $\{e_1, e_2, e_3\}$ is a basis and **$\dim(\mathbb{R}^3) = 3$** .

Q2 (Easy): Show that $B = \{(1, 0), (0, 1)\}$ is a basis of $\mathbb{R}^2$.

Solution:

Spanning: $(x, y) = x(1,0) + y(0,1)$.

Independence: $a(1,0) + b(0,1) = (a, b) = 0 \Rightarrow a = b = 0$.

Both conditions hold, so **B is a basis** and $\dim(\mathbb{R}^2) = 2$.

★★ Medium

Q3 (Medium): Show that $B = \{(1,1,0), (1,0,1), (0,1,1)\}$ is a basis of $\mathbb{R}^3$.

Solution:

Strategy: $\dim(\mathbb{R}^3) = 3$ and B has 3 vectors — it suffices to prove independence.

$a(1,1,0) + b(1,0,1) + c(0,1,1) = 0$ gives:

$a + b = 0, \quad a + c = 0, \quad b + c = 0.$

Step 1: $b = -a, c = -a$. Sub into $b + c = 0$: $-2a = 0 \Rightarrow a = 0$. Then $b = c = 0$.

Independent + size 3 in $\mathbb{R}^3 \Rightarrow$ **B is a basis of $\mathbb{R}^3$.**

Alternate Solution — Row Reduce to REF:

Form the matrix with the three vectors as rows and reduce:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_1$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$R_3 \rightarrow R_3 + R_2$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Three non-zero rows $\Rightarrow$ rank = 3. Combined with 3 vectors in a 3-dim space, **B is a basis of $\mathbb{R}^3$.** ✓

Equivalent shortcut: det of the matrix with these rows is $-2 \neq 0$.

Q4 (Medium): Find a basis and the dimension of $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.

Solution:

Strategy: express x in terms of free variables y and z .

From $x + y + z = 0$: $x = -y - z$. So a generic element is $(x, y, z) = (-y - z, y, z) = y \cdot (-1, 1, 0) + z \cdot (-1, 0, 1)$.

Let $v_1 = (-1, 1, 0), v_2 = (-1, 0, 1)$. They span W .

Independence: $a \cdot v_1 + b \cdot v_2 = (-a-b, a, b) = (0,0,0) \Rightarrow a = 0, b = 0$.

So $\{v_1, v_2\}$ is a basis and **$\dim(W) = 2$** . (W is a plane through the origin.)

★★★ Difficult

Q5 (Difficult): Find a basis and the dimension of $W = \{(x, y, z, w) \in \mathbb{R}^4 : x + y = 0 \text{ and } z - w = 0\}$.

Solution:

Strategy: two independent constraints in $\mathbb{R}^4$ should leave $4 - 2 = 2$ free directions.

From $x + y = 0$: $x = -y$. From $z - w = 0$: $z = w$. Pick y and w as parameters.

Generic element: $(x, y, z, w) = (-y, y, w, w) = y \cdot (-1, 1, 0, 0) + w \cdot (0, 0, 1, 1)$.

Let $v_1 = (-1, 1, 0, 0)$, $v_2 = (0, 0, 1, 1)$.

Independence: $a \cdot v_1 + b \cdot v_2 = (-a, a, b, b) = 0$ forces $a = b = 0$.

Hence $\{v_1, v_2\}$ is a basis and **$\dim(W) = 2$** .

Q6 (Difficult): Find the dimension and a basis of the vector space S of all 2×2 symmetric matrices.

Solution:

Strategy: count free entries. A symmetric 2×2 matrix has the diagonal entries and one off-diagonal — three free numbers.

General form: $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ with $a, b, c \in \mathbb{R}$. Decompose:

$A = a \cdot E_1 + b \cdot E_2 + c \cdot E_3$, where $E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $E_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $E_3 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$.

Spanning: every symmetric A is built from $\{E_1, E_2, E_3\}$.

Independence: $\alpha \cdot E_1 + \beta \cdot E_2 + \gamma \cdot E_3 = \begin{bmatrix} \alpha & \beta \\ \beta & \gamma \end{bmatrix} = 0$ forces $\alpha = \beta = \gamma = 0$.

Hence $\{E_1, E_2, E_3\}$ is a basis and **$\dim(S) = 3$** .

Compare with all 2×2 matrices: $\dim = 4$. Symmetry kills one degree of freedom.

Review at a Glance

Concept	Quick test
Subspace $W \subseteq V$	$0 \in W$; closed under $+$ and scalar $\cdot$
v is a linear combination	Solve $c_1 v_1 + \dots + c_k v_k = v$ for scalars
$v_1, \dots, v_k$ independent	$c_1 v_1 + \dots + c_k v_k = 0 \Rightarrow$ all $c_i = 0$
Basis	Independent AND spans the space
Dimension	Number of vectors in any basis

Tips for Students

- ✓ **Start with the picture.** Before any algebra, ask: is this a line, a plane, all of $\mathbb{R}^n$, or a curved/shifted set? Linear means flat; subspaces always include the origin.
- ✓ **Always verify the zero vector first.** If 0 isn't in your set, you're done — it can't be a subspace. This 5-second check often kills the problem instantly.
- ✓ **Translate words into a system.** Independence and span questions become homogeneous systems. The shape of the solution set tells you the answer: only zero $\Rightarrow$ independent / spanning fails; free variables $\Rightarrow$ dependent / extra room.

- ✓ **Use the size shortcut.** In $\mathbb{R}^n$, any n linearly independent vectors automatically form a basis — no need to recheck spanning. Likewise, any n vectors that span $\mathbb{R}^n$ are automatically independent.
- ✓ **Sanity-check with determinants.** For n vectors in $\mathbb{R}^n$, $\det = 0 \Leftrightarrow$ dependent. It's the fastest test once you're comfortable.
- ✓ **Count free variables for dimension.** Dimension of a subspace defined by k independent linear equations in $\mathbb{R}^n$ is $n - k$ (the Rank-Nullity intuition). Use this to predict your answer before solving.

End of worksheet. *Encourage students to attempt each problem first, then read the solution to identify gaps. Concept boxes (definition, intuition, strategy, watch-out) are designed to be read repeatedly — they distill the heart of each topic.*